

Elevator Pitch

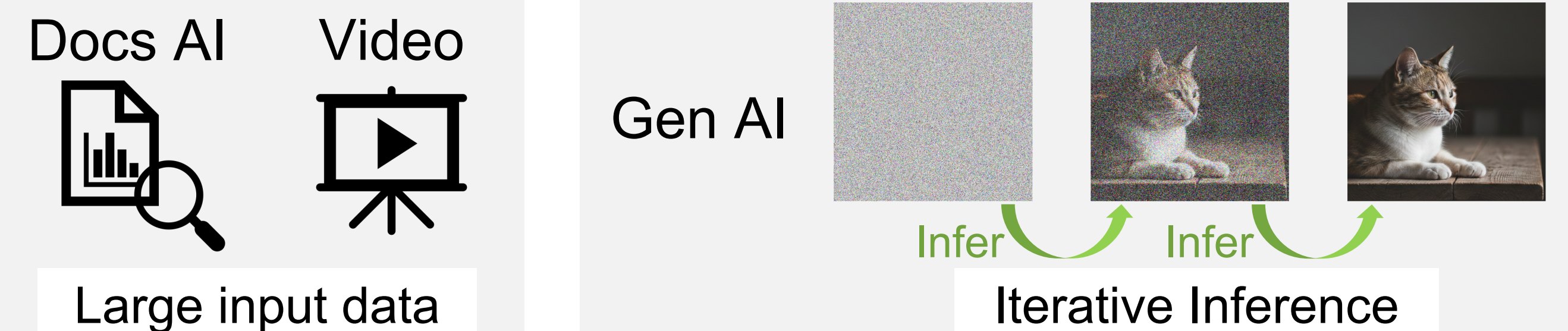
Discovery 🔍: Token Merging for faster ViTs **distorts attention** 😞

Proposal 💡: Lightweight attention score correction *ViT: Vision Transformer

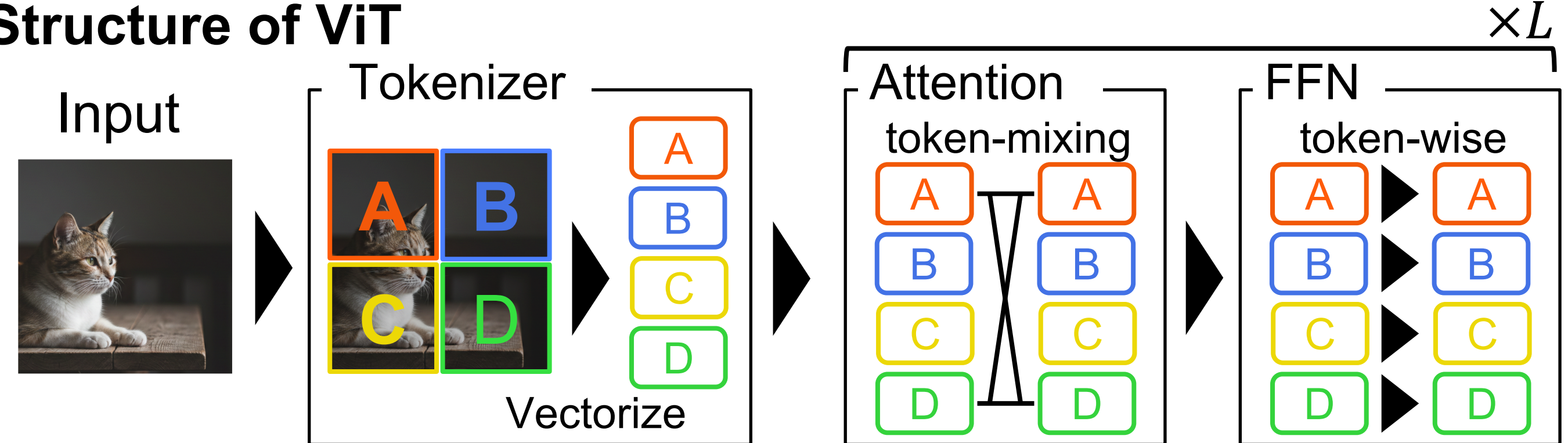
Achievement 🏆: Better speed-accuracy trade-off 😊

Background

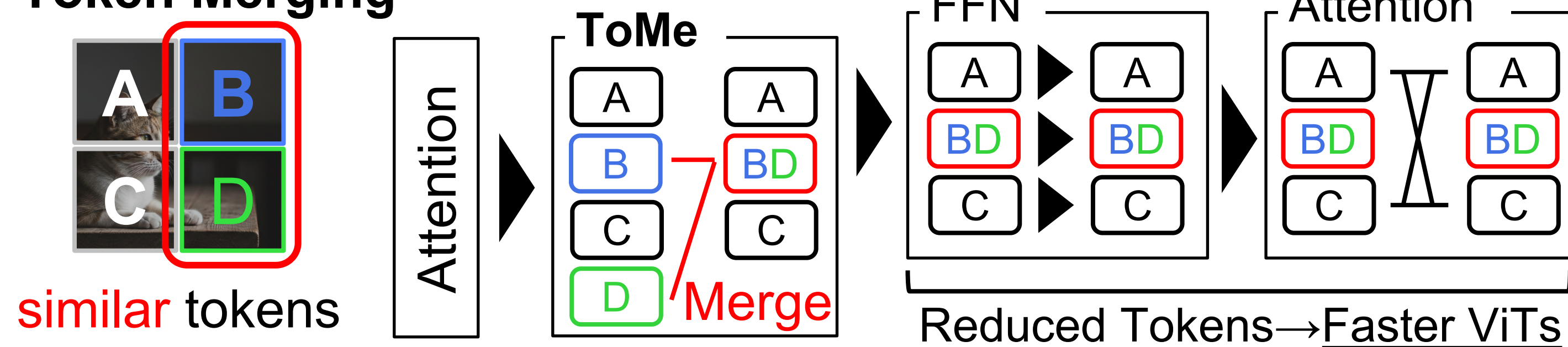
Motivation: Why Faster ViTs?



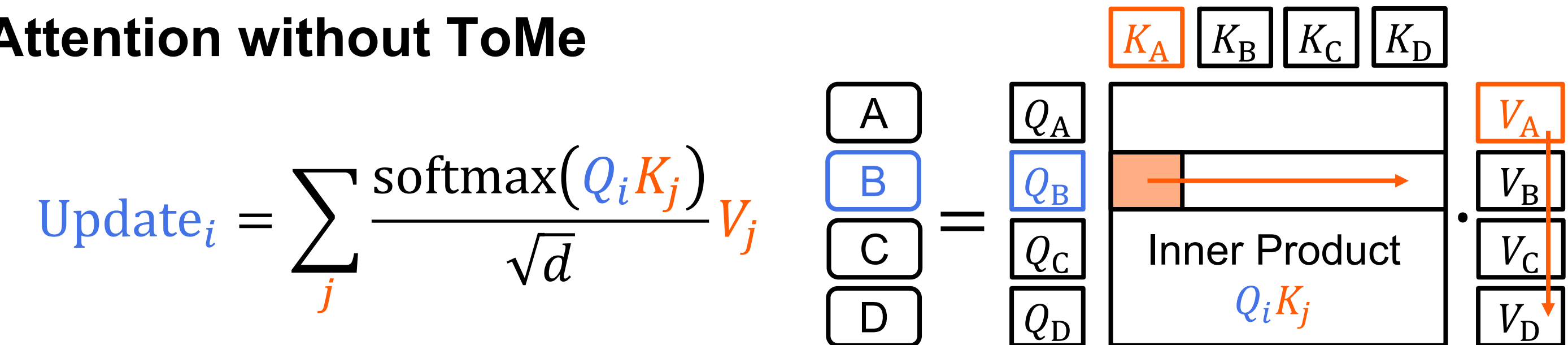
Structure of ViT



Token Merging

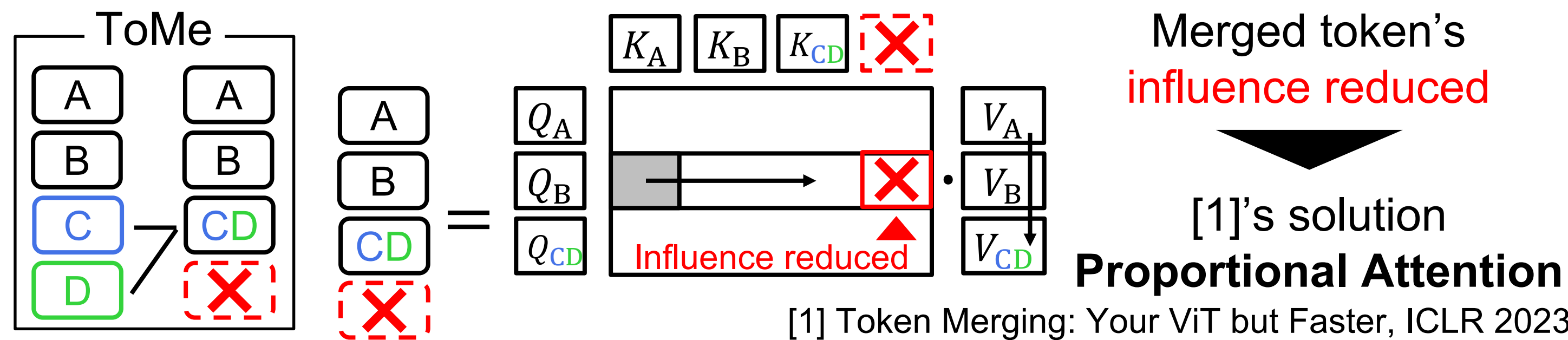


Attention without ToMe

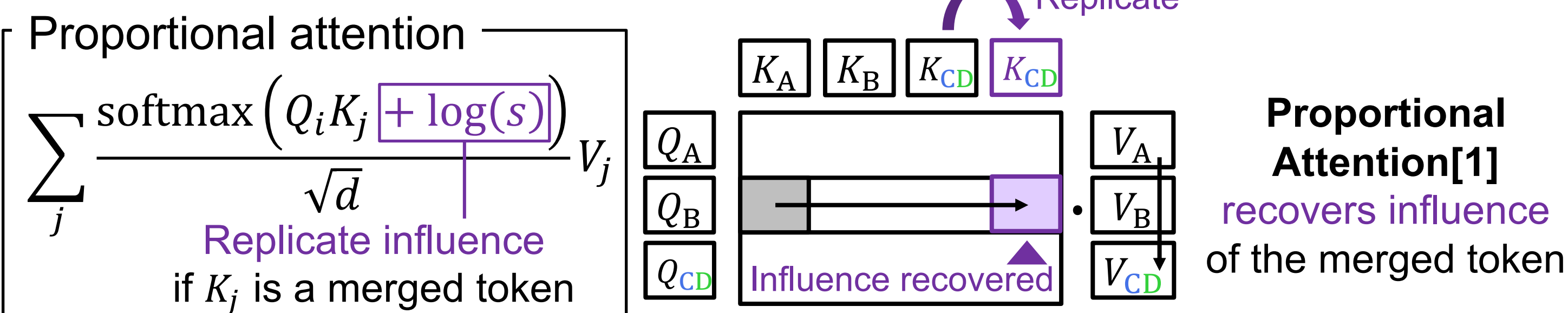


Discovery

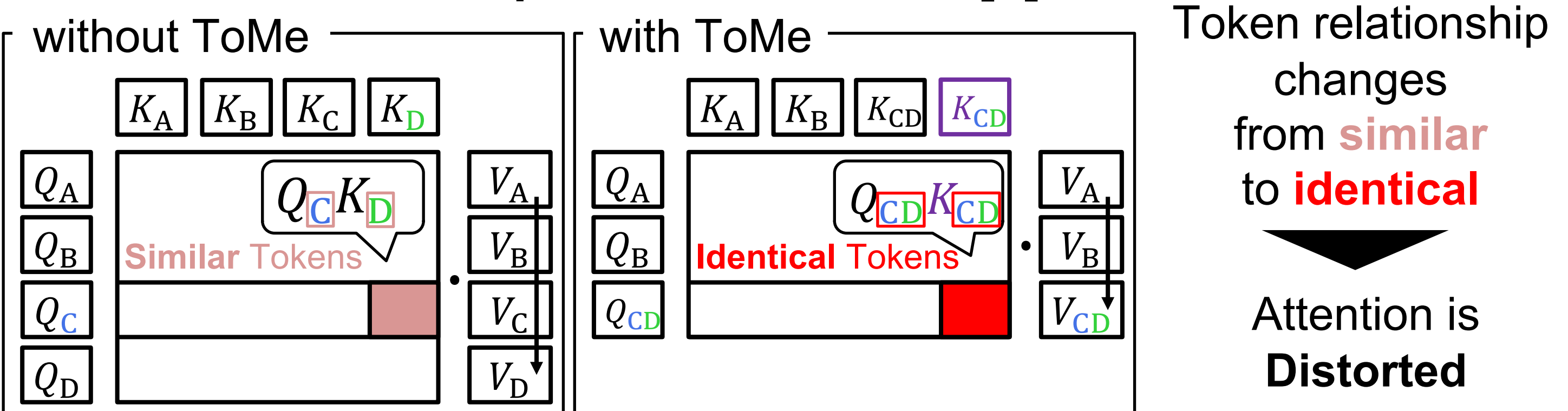
Attention with naïve ToMe (merging C and D)



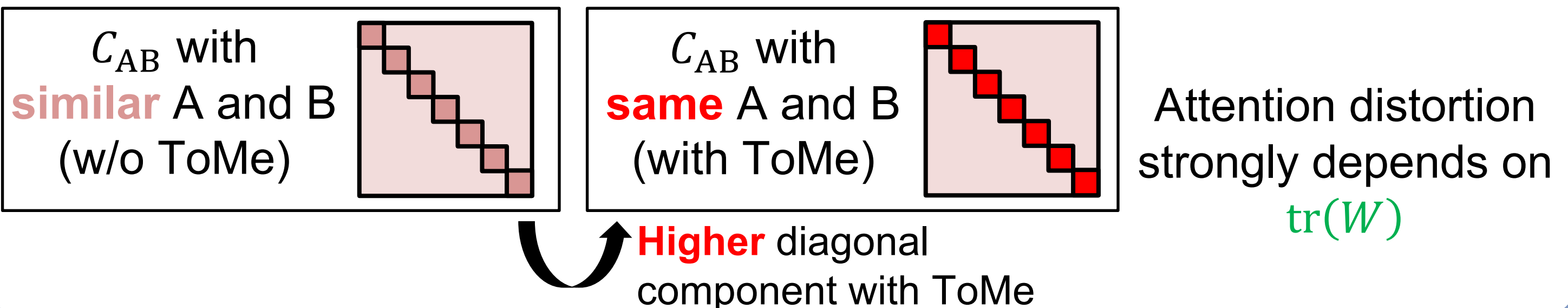
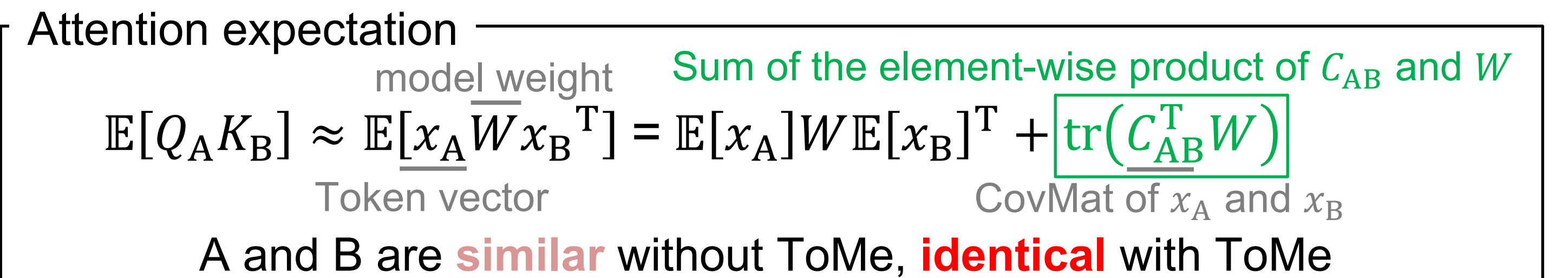
Attention with ToMe + PropAttn[1]



Hidden Issue in Proportional Attention[1]



How does Attention Distort?



Proposal

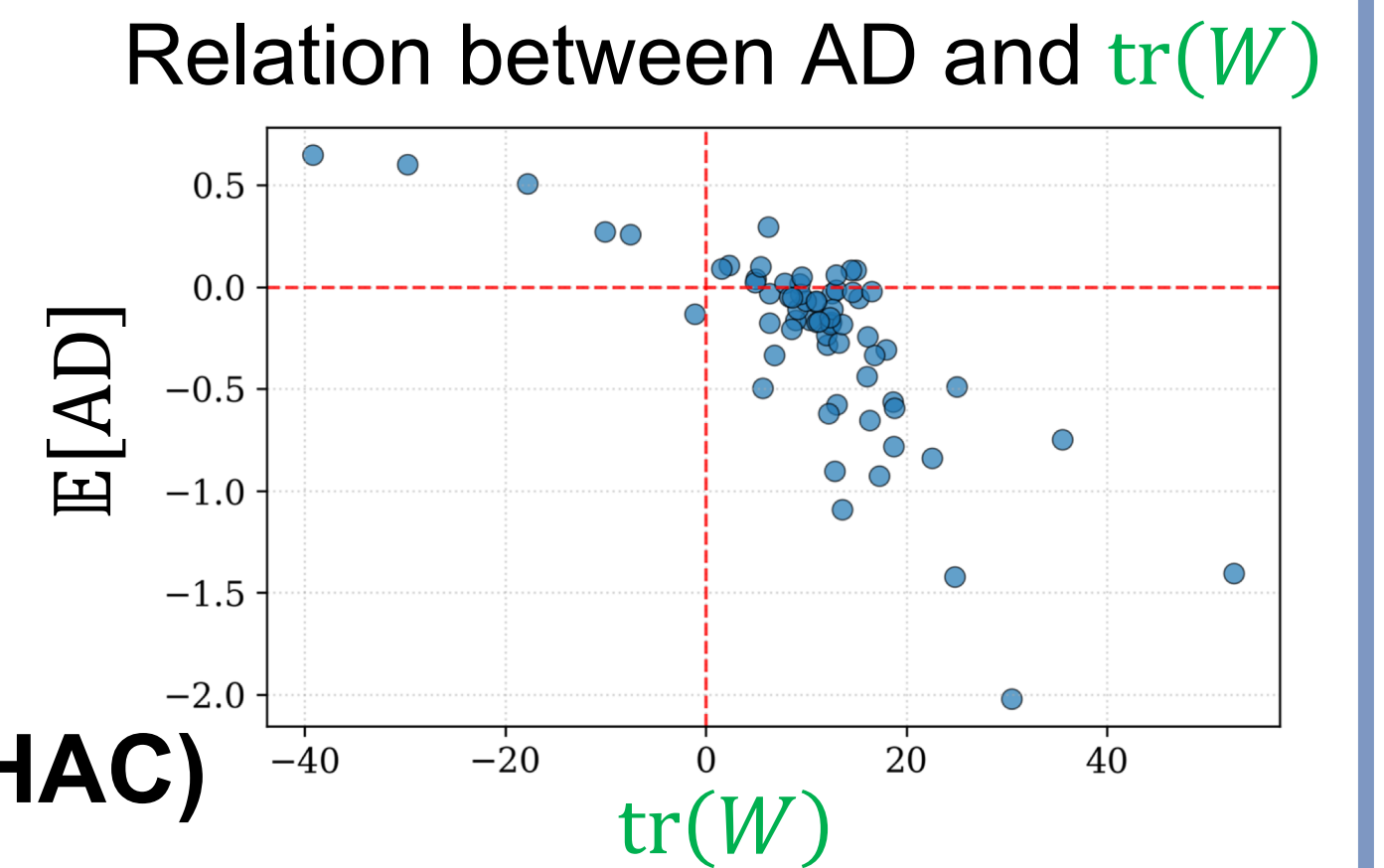
Quantifying Attention Distortion

We define Attention Distortion (AD) = $Q_A K_B - Q_{\text{Merged}} K_{\text{Merged}}$
(w/o ToMe, Ideal) (with ToMe, Distorted)

Right: Empirical Validation on DeiT-S with ToMe ($r=10$), ImageNet Validation set

AD is $\text{tr}(W)$ -dependent
(Its trends vary by head)

We propose
Head-Wise Attention Correction (HAC)



Statistical-HAC (S-HAC)

Making attention close to Ideal
 $Q_i K_j + \log(s) + \mathbb{E}[AD]$ if j is Merged
Distorted Ideal - Distorted

Learned-HAC (L-HAC)

Lightweight alternative
 $Q_i K_j + \log(s) + f(s)$
Learned correction

Achievement

Evaluation of S-HAC

Model	Method	$r = 20$	
		Acc (%)	Thr (im/s)
DeiT-B	Full Model	81.8	1400
	No Corr	68.3	3031
	Prop Attn	66.7	2975
	S-HAC	68.8 +2.1%	2965 slow
DeiT-S	Full Model	79.8	3865
	No Corr	64.7	7982
	Prop Attn	74.1	7803
	S-HAC	74.5 +0.4%	7457 slow
DeiT-T	Full Model	72.1	8947
	No Corr	46.3	16293
	Prop Attn	61.1	16282
	S-HAC	61.7 +0.6%	15537 slow
S-HAC		improves accuracy 😊 suffers from overhead 😞	

Evaluation of L-HAC

